

Arvinth Shanmugam

Post-Baccalaureate Research Fellow, WSAI, IIT Madras

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Education

July 2025	Indian Institute of Technology Madras Integrated B.S. & M.S. in Biological Sciences, Specialization in Computational Biology	GPA: 8.77/10
Aug 2020		Class Rank: 4 Last 2 years : 9.53/10

Research Experience

Present Jul 2025	Post Baccalaureate Researcher Wadhvani School of Data Science & AI 🌐 IITM, Chennai, India Advisors: Prof. Nirav Bhatt , Prof. Ganesh Ramakrishnan Research Topics: Spatiotemporal Model [C3], ML for Sparse Data [S1, W2], Cost Effectiveness Analysis [S2] > Recipient of a merit-based research fellowship (\$600/month), awarded to exceptional graduates
June 2025 July 2024	Master's Research Thesis National Disease Modeling Consortium 🌐 IITB, Bombay, India Advisors: Prof. Nirav Bhatt , Prof. Ganesh Ramakrishnan Research Topics: Intervention Impact Modeling & Optimal Control [J3,C2, P3,P2] > Nominee for Batch of 1979 Award for the best Dual Degree Project Thesis in Biological Sciences
Feb 2025 Dec 2024	Undergraduate Research Thesis Centre for Public Health Kinetics 🌐 Remote Advisors: Dr.Sunil Sazawal , Prof. Nirav Bhatt Research Topics: Non-Linear Mixed Effects Modeling [J1], Multi-State Markov Modeling [J2]
July 2024 May 2024	Research Intern Artificial Intelligence and Robotics Laboratory 🌐 IISc, Bangalore, India Advisor: Prof. Suresh Sundaram Research Topics: Vehicular Modeling, RBF-Based Neural Network, Velocity Control
April 2023 Sept 2021	Undergraduate Researcher Team Avishkar 🌐 IITM, Chennai, India Advisors: Prof. Satya Chakravarthy , Prof. Muruganandam T.M Research Topics: Non-Linear Newtonian Modeling, Parameter and State Estimation, Adaptive Control [C1]

Publications

J = Journal

- [J.3] **Modeling the Impact of Active HPV Testing Intervention on Cervical Cancer Burden in India** 🌐
S Arvinth, Rashmi Tiwari, S Vishnu Tej, Piyush Sawarkar, Prerna Bhalla, Shekar Sivasubramanian, Ajit Rajwade, Ganesh Ramakrishnan[✉], Nirav Bhatt[✉]
Infectious Disease Modeling Journal (Under Revision), [medRxiv](#) [IDM'2025]
- [J.2] **Markov Modelling for Estimating Transition Probabilities for Malnutrition During the First Two Years of Life**
Nirav Bhatt[†], S Arvinth[†], Rasheda Khanam[†], Naveed Iqbal[†], Sayan Das, Md. Shafiqul Islam, Waqasuddin Khan, Usha Dhingra, Salahuddin Ahmed, Farzana Bashir, Tulika Sinha, Nabilul Haque Chowdhury, Farrukh Qazi, Arup Dutta, Arunangshu Dutta Roy, Samiah Kanwer, Saikat Deb, Dipak Mitra, Salman Usmani, Innocent Nyalala, Diwakar Mohan, Muhammad Imran Nisar, Rajiv Bahl, Sachiyo Yoshida, Fyezah Jehan[✉], Abdullah H. Baqui[✉], Sunil Sazawal[✉]
Journal of Global Health (Under Revision) [JoGH'2025]
- [J.1] **Understanding Child Growth Patterns in the First Two Years of Life: A SITAR Analysis of AMANHI-ACT Cohorts**
Sunil Sazawal^{†✉}, Nirav Bhatt[†], Rasheda Khanam[†], Naveed Iqbal[†], S Arvinth, Salahuddin Ahmed, Samiah Kanwar, Sayan Das, Abrarul Haque Asif, Farrukh Qazi, Arup Dutta, Md. Shafiqul Islam, Waqasuddin Khan, Pavan Kumar S, Nabilul Haque Chowdhury, Najeeb Rehman, Saikat Deb, Dipak Mitra, Farzana Bashir, Mayassa Ally, Diwakar Mohan, Fyezah Jehan, Rajiv Bahl, Sachiyo Yoshida, Muhammad Imran Nisar, Abdullah H. Baqui, Usha Dhingra
Journal of Global Health (Under Revision) [JoGH'2025]

Working Papers

S = Working Paper

- [S.2] **Structured Opportunistic Prostate Cancer Screening in Urban India: A Modelling Study for Low- and Middle-Income Countries**
Srinivas Chilukuri[✉], Sham Sundar, S Arvinth, Nirav Bhatt
Working Paper
- [S.1] **A machine-learning framework for global HPV prevalence prediction in data-limited settings**
Prerna Bhalla[†], S Arvinth[†], Sainath Raja, Daljeet Singh, Nirav Bhatt[✉]
Working Paper

- [C.3] **Modeling the State-Wise Impact of Coordinated HPV Vaccination and Screening on Cervical-Cancer Outcomes in India**
S Arvinth, Rashmi Tiwari, Prerna Bhalla, Ajit Rajwade, Ganesh Ramakrishnan, Nirav Bhatt[✉]
International Conference on Modelling and Health Informatics for Public Health (Accepted) [ICMIPH'25]
- [C.2] **Optimal Active HPV screening strategy for reducing HPV infections and Cervical Cancer in Developing Countries**
Rashmi Tiwari[†], S Arvinth[†], S Vishnu Tej, Piyush Sawarkar, Prerna Bhalla, Shekar Sivasubramanian, Ajit Rajwade, Ganesh Ramakrishnan[✉], Nirav Bhatt[✉]
Infectious Disease Modeling Conference 2024, Bangkok, Thailand [IDM'24]
- [W.2] **Develop Machine Learning Models to predict HPV prevalence using Meta-Data** [✉]
Nirav Bhatt, Prerna Bhalla, S Arvinth, Sainath Raja, Daljeet Singh
Workshop on Core Modeling Frameworks and Open-Source Tools by the Institute of Disease Modeling (IDM)
- [P.3 , P.2] **HPV Screening Strategy for Reducing HPV Infections and Cervical Cancer using Optimal Control** [✉]
S Arvinth, Rashmi Tiwari, S Vishnu Tej, Piyush Sawarkar, Prerna Bhalla, Shekar Sivasubramanian, Ajit Rajwade, Ganesh Ramakrishnan[✉], Nirav Bhatt[✉]
WSAI Annual Research Showcase 2025 & AI/ML in Process Systems Engineering 2025 [WSAI-ARS'25, AI/ML in PSE]
- [P.1] **Accelerated-Gradient-Based Flexible-Object Transport With Decentralized Robot Teams** [✉]
S Arvinth, Abdullah
Project Presentation, Convex Optimization
- [C.1] **A Decentralized EMS Guidance System for Hyperloop: Architecture, Cascaded Control, and Experimental Validation** [✉]
S Arvinth, Karthik Shenoy, Jay Gupta, Sumanth, Rudra Panch, Pranav Debbad
European Hyperloop Week 2023, The University of Edinburgh, Scotland [EHW'23]

Workshop Organization

- [W.1] **Workshop on Role of Technology for Advancing Women's Health in India**
Nirav Bhatt, Prerna Bhalla, S Arvinth, Akash Chakraborty
Speakers: Prof. Krithika Rangarajan, Prof. Vijayalakshmi Ramshankar, Prof. Rupsa Bhattacharjee, Dr. Oshima Sachin et. al.

Select Research Projects

- Modeling and Control of Epidemiological Processes for HPV Spread and Cervical Cancer** NDMC IITB
Advisor: Prof. Nirav Bhatt
- > Developed and calibrated age-structured compartmental models for HPV transmission and cervical carcinogenesis, aligned with Indian epidemiological baselines.
 - > Extended the framework to integrate age-targeted HPV vaccination and population screening, enabling impact and evaluation of combined intervention policy scenarios.
 - > Conducted simulations quantifying the effects of varying vaccination/screening coverage, target age bands, and introduction timing on infections, precancerous lesions, cancer cases, and mortality.
 - > Generated State-Level Population Projections data using Bayesian hierarchical modeling & Logistic-window functions
 - > Built a decision analytic framework by solving an optimal-control problem to allocate time-varying vaccination and screening intensities under resource constraints, minimizing long-run cervical-cancer burden.
- Predicting Global HPV and CIN Burden from Limited and Missing Data using Machine Learning** NDMC IITB
Advisor: Prof. Nirav Bhatt
- > Curated multi-country epidemiologic, sociodemographic & health system covariates; engineered STI-burden and disease-incidence indices; engineered composite risk indices and applied imputation to address missingness.
 - > Built and cross-validated machine-learning models on imbalanced targets using data rebalancing using SMOGN.
 - > Predicted the Global and Indian state-level disease burden estimates with diagnostic analyses.
- Agent-Based Modeling with HPVsim for State-Level HPV/Cervical-Cancer Dynamics** NDMC IITB
Advisor: Prof. Nirav Bhatt
- > Extended HPVsim with maintained patches to enable subnational stratification with state-level demographic and epidemiological inputs
 - > Performed state-level microsimulations for India incorporating variation in sexual behavior and demographic patterns.

Longitudinal Growth and Malnutrition Dynamics in Early Childhood: AMANHI-ACT Cohorts

CPHK

Advisors: *Dr. Sunil Sazawal, Prof. Nirav Bhatt*

- **SITAR Growth-Curve Modeling (Bangladesh, Pakistan, Tanzania).**
 - Fitted nonlinear mixed-effects **SITAR** models for height and weight, extracting size, timing, and intensity parameters with subgroup stratification (sex, SGA/AGA, birth-weight classes).
 - Generated mean and velocity curves and random-effects correlation maps to characterize cohort-specific growth trajectories and identify early growth faltering signals.
- **Multi-State Markov Modeling of Malnutrition Transitions.**
 - Defined seven nutritional states (Normal; Underweight, Stunted, Wasted; and compounded states) and estimated continuous-time transition intensities and one-year transition probabilities from cohort follow-ups.
 - Quantified Median First Passage/Recurrence Times and lifetime time-in-state to reveal persistence of stunting and rapid cycling among compounded deficits, versus relatively faster recovery from wasting.

Cost-Effectiveness Analysis using Markov Modeling for Opportunistic Prostate Cancer Screening

Apollo

Advisors: *Dr. Srinivas Chilukuri, Prof. Nirav Bhatt*

- Developed a discrete-time cohort Markov model of prostate-cancer natural history and screening pathways, parameterized with urban Indian epidemiology over a 20-year horizon.
- Performed cost-effectiveness analysis from a health-system perspective, estimating costs and QALYs and reporting the incremental cost-effectiveness ratio (ICER) versus no screening.
- Conducted uncertainty analysis (deterministic and probabilistic) and scenario testing to assess robustness of clinical and economic outcomes.

Talks

“Levitation and Guidance for Hyperloop Pods”

- European Hyperloop Week 2023 - University of Edinburgh, Scotland July 2023 (Edinburgh, UK)

“Propulsion Control and Regenerative Braking for Hyperloop”

- European Hyperloop Week 2022 - TU Delft, Netherlands July 2022 (Delft, NL)

Honours and Awards

- **Post Baccalaureate Fellowship, Wadhvani School of Data Science & AI** — Merit-based research fellowship (\$600/month).
- **WSAI Travel Grant, 2025** — For attending ICMIPH’25, IIT Bombay, India.
- **Ministry of Railways Travel Grant (EHW 2023)** Competitive travel award (\$1,200) to attend EHW 2023, Scotland.
- **Ministry of Railways Travel Grant (EHW 2022)** Competitive travel award (\$1,200) to attend EHW 2022, Netherlands.
- **Amazon ML Challenge, 2024** — Team achieved **All-India Rank 130**.
- **Dayanand Academic Excellence Award, School Topper** — Secured **493/500** in the Grade X Board Exam.

Teaching and Positions of Responsibility

- **Teaching Assistant** — Department of Biotechnology, IIT Madras
 - **Life Sciences** Jul–Nov 2024
 - Mentored and assisted a cohort of 100+ students; designed assignments, homework problems, and project briefs.
 - Organized and evaluated a team-presentation project (24 teams) using structured rubrics and written feedback.
 - **Genomics and Proteomics** Jan–May 2025
 - Supported a class of 50+ students; Authored and evaluated class assignment problems.
- **Volunteer Teacher & Mentor, Avanti Fellows** — Puducherry, India [🌐] Aug 2021–April 2022
 - Taught Physics, Chemistry, and Math at JEE level, focusing on personalized support for under-resourced students.
 - Designed and evaluated full-length mock exams and weekly assignments; conducted feedback-driven review sessions.
 - Mentored and tracked the progress of 40 students throughout the year; 12 secured admission to IITs and NITs.
- **Lead Chief Engineer, Team Avishkar Hyperloop** — CFI, IIT Madras [🌐] Sep 2021–Apr 2023
 - Led a 72-member team across seven subsystems; oversaw technical architecture, integration, and validation; planned and executed major technical decisions.

Key Courses Undertaken

Quantitative	Biostatistics, Linear Algebra, Convex Optimization, Multivariate Calculus, Series & Matrices, Numerical Methods, Theoretical Biophysics
CS & Economics	Data Structures and Algorithms, Bioinformatics, Statistical Inference, Principle of Economics, Climate Economics, Computational Neuroscience, Computational Biology Lab
ML & Control	Pattern Recognition and Machine Learning, Modern Control Theory, Intro to Robotics